

PROJECT DESIGN PHASE

Solution Architecture

Date	
Team ID	
Project Name	Auto-Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Solution Architecture

The solution architecture describes how the business requirement for simpler school IT ticket submission is connected to the ServiceNow technology components. The design separates the user interface, ticket data, automation logic, notification service, and controlled configuration.

Architecture Objectives

- Provide a simple entry point for students to submit an IT request.
- Store requests in a dedicated School IT Ticket table.
- Automatically classify supported issues using Flow Designer rules.
- Notify the caller after the request is received.
- Maintain security and controlled changes through ServiceNow roles, ACLs, and Update Sets.

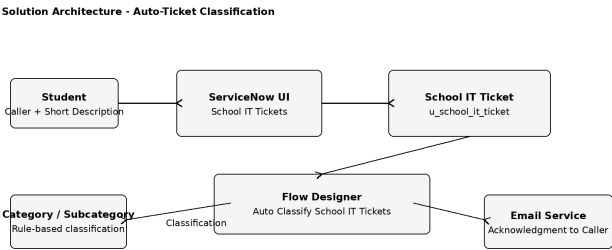


Figure 1: Solution architecture and data flow of the School IT ticket classification system.

Architecture Flowchart & Diagram Explanation

System Flow & Layer Interaction:

- **Presentation Layer:** Students interact with the ServiceNow Next Experience School IT Ticket form. They input only their identity (Caller) and issue summary (Short Description).
- **Data Persistence Layer:** The custom table `u_school_it_ticket` captures the record and triggers database events.
- **Automation Layer:** Flow Designer listens asynchronously for record creation, inspects text fields, evaluates conditional keyword rules, and updates dependent fields.
- **Communication Layer:** The platform email engine formats and dispatches the acknowledgment message directly to the requester.

Architecture Components

Component	Role in the solution	Technology
User Interface	Students enter Caller and Short Description and access School IT Tickets.	ServiceNow Next Experience
Ticket Data	Stores Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group and Assigned To.	Custom table: u_school_it_ticket
Automation Logic	Checks the Short Description and updates Category and Subcategory for supported issue types.	ServiceNow Flow Designer
Notification	Sends an acknowledgment email to the Caller after ticket submission.	ServiceNow Email / SMTP
Security	Controls access to the custom table and its records.	ServiceNow ACLs and Roles
Change Management	Captures configuration changes for deployment and auditability.	ServiceNow Update Sets

Implementation Evidence

The screenshot displays the 'School IT Ticket' form in the ServiceNow interface. The form includes the following fields:

- Number:** A text field with the value 'TKT0000012'.
- Caller:** A dropdown menu showing 'System Administrator'.
- Short description:** A text area with the value 'Wi-Fi not working in library'.
- Description:** A large text area for additional details.
- Category:** A dropdown menu showing 'Network'.
- Assigned Group:** A dropdown menu with a search icon.
- Subcategory:** A dropdown menu showing 'Wi-Fi'.
- Assigned To:** A dropdown menu with a search icon.
- State:** A dropdown menu showing 'None'.

At the bottom of the form, there are 'Update' and 'Delete' buttons.

Figure 2: School IT Ticket form showing the user-facing ticket submission fields.

Implementation Evidence - Automation

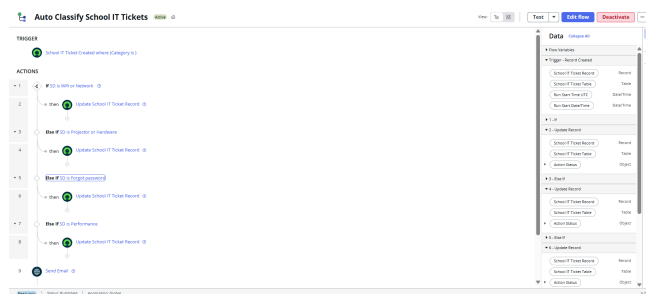


Figure 3: Published Auto Classify School IT Tickets Flow Designer workflow.

The screenshot shows the 'School IT Ticket' form in ServiceNow. The ticket number is 'TK1001012'. The category is 'Network'. The status is 'Open'. The assigned user is 'System Administrator'. The form includes fields for 'Number', 'Category', 'Status', 'Assigned To', and 'Updated'. The 'Update' button is visible at the bottom.

Figure 4: School IT Ticket showing the result of automatic classification.

The screenshot shows the 'Update Set: Project Update Set' in ServiceNow. It displays a list of updates with columns for 'Created', 'Type', 'View', 'Target scope', 'Update type', 'Source update set', and 'Action'. The table contains several rows of update records, including 'Project Update Set', 'Module', 'Classified', 'Classified', 'Classified', 'Classified', 'Classified', and 'Classified'. The 'Update' button is visible at the bottom.

Created	Type	View	Target scope	Update type	Source update set	Action
2020-09-28 10:10:10	Project Update Set		School IT Ticket	action	Import	Import
2020-09-28 10:10:10	Module		School IT Ticket	action	Import	Import
2020-09-28 10:10:10	Classified		School IT Ticket Category	action	Import	Import
2020-09-28 10:10:10	Classified		School IT Ticket Status	action	Import	Import
2020-09-28 10:10:10	Classified		School IT Ticket Subcategory	action	Import	Import
2020-09-28 10:10:10	Table		School IT Ticket	action	Import	Import
2020-09-28 10:10:10	Database		School IT Ticket	action	Import	Import
2020-09-28 10:10:10	Database		School IT Ticket Answer Group	action	Import	Import

Figure 5: Completed Project Update Set containing the captured customizations and XML export option.